

MealFlex Mobile Application Outline Report

I. Project Description

A. Overview of MealFlex

1. MealFlex is an Android mobile application designed to assist individuals and families in organizing weekly meals while managing grocery budgets effectively.
2. The application integrates meal planning, recipe management, grocery list generation, and budget tracking into a single, user-friendly mobile platform.
3. MealFlex is designed to simplify daily food decision-making and improve household efficiency.

B. Purpose of the Application

1. To help users organize recipes and create structured weekly meal plans.
2. To automatically generate grocery lists based on selected meals and ingredients.
3. To reduce food waste by improving planning and ingredient tracking.
4. To support users in lowering grocery expenses through budget-conscious meal suggestions and planning tools.

II. Problem Addressing

A. Rising Grocery Costs and Inflation

1. Grocery prices continue to increase due to inflation and supply chain challenges.
2. Many families struggle to maintain consistent food budgets while meeting nutritional needs.

B. Difficulty Managing Meal Planning with Busy Schedules

1. Individuals and families often lack time to plan meals in advance.
2. This leads to last-minute food decisions that are often less healthy or more expensive.

C. Food Waste Caused by Poor Organization

1. Without proper planning, ingredients are often purchased but not fully used.
2. This results in expired food and unnecessary financial loss.

D. Lack of Affordable Meal Planning Tools

1. Many existing meal planning applications require paid subscriptions or offer limited features.
2. MealFlex provides a free, accessible, and simplified alternative focused on essential household needs.

III. Platform**A. Mobile Platform**

1. The application will be developed for the Android operating system.
2. It will be compatible with both Android smartphones and tablets.

B. Development Environment and Tools

1. MIT App Inventor will be used for initial prototyping and interface development.
2. Android Studio will support advanced development and future expansion.
3. Visual Studio Code will be used for supporting code development and documentation.
4. GitHub will be used for version control, collaboration, and project documentation.

IV. Front-End and Back-End Support**A. Front-End (User Interface)**

1. The application will feature a clean and intuitive Android user interface.
2. Navigation will include clearly labeled sections such as Home, Meal Planner, Recipes, Grocery List, and Budget Tracker.

3. The design will be responsive to different screen sizes and optimized for mobile usability.
4. The interface will prioritize simplicity to ensure ease of use for all users.

B. Back-End (System and Data Management)

1. Firebase Authentication will be used to securely manage user login and account creation.
2. Firebase Firestore will store user data such as meal plans, recipes, and grocery lists.
3. Cloud-based storage will allow real-time synchronization across multiple devices.
4. The system will support offline storage to allow limited functionality without internet access.
5. External recipe APIs may be integrated to expand recipe availability and variety.

V. Functionality

A. User Login

1. Users will be able to create secure accounts and log into the application.
2. Firebase Authentication will ensure that all user data is protected and accessible only to authorized users.

B. Weekly Meal Planner

1. Users can create structured weekly meal plans using a calendar-based interface.
2. Meals can be assigned to specific days and categorized by meal type (breakfast, lunch, dinner).

C. Recipe Organization

1. Users can store and manage a personal collection of recipes within the application.
2. Each recipe will include ingredients, preparation steps, and optional nutritional information.

D. Grocery List Generator

1. The application will automatically generate grocery lists based on selected meal plans.

2. Ingredients will be organized into categories to improve shopping efficiency and reduce time spent in stores.

E. Budget Tracking

1. Users can set weekly or monthly grocery budgets within the application.
2. The system will estimate spending based on selected meals and ingredients to help users stay within budget.

F. Meal Rotation Suggestions

1. The application will suggest previously used meals to reduce repetition fatigue.
2. This feature will help users maintain variety in their weekly meal planning.

G. Cloud Synchronization

1. All user data, including meal plans, grocery lists, and recipes, will be stored in the cloud.
2. Firebase will enable real-time synchronization across multiple devices.

H. Offline Access

1. Users will be able to access saved meal plans and grocery lists without an internet connection.
2. Any changes made offline will automatically sync once the device reconnects to the internet.

VI. Design (Wireframes)

A. Login Screen

1. Provides secure user authentication through email and password login.

B. Home Dashboard

1. Serves as the main navigation hub of the application.
2. Displays weekly summary information and quick access to key features.

C. Meal Planner Screen

1. Allows users to assign meals to specific days of the week.

2. Provides a visual calendar-based planning interface.

D. Grocery List Screen

1. Displays automatically generated shopping lists based on meal selections.
2. Items are grouped by category for easier shopping organization.

E. Recipe Details Page

1. Displays full recipe information including ingredients and instructions.
2. May include nutritional data and preparation time.

F. Budget Tracking Section

1. Displays current spending compared to the user's set budget.
2. Provides visual indicators to help users manage grocery expenses effectively.

VII. Development Goals

1. To design and develop a fully functional Android mobile application.
2. To create a clean, intuitive, and user-friendly interface.
3. To integrate Firebase services for authentication and cloud storage.
4. To demonstrate proper use of GitHub for version control and documentation.
5. To deliver a professional presentation and live demonstration of the application.

VIII. Conclusion

MealFlex is a mobile application designed to simplify meal planning, reduce food waste, and help users manage grocery expenses more effectively. By combining meal organization, recipe management, grocery list automation, and budget tracking into a single Android platform, MealFlex provides a practical and accessible solution for busy individuals and families seeking to improve their daily food planning and financial efficiency.

README Link:

<https://github.com/Kennisha-Commits/MealFlex/blob/main/README.md>

Wiki Link:

<https://github.com/Kennisha-Commits/MealFlex/wiki/MealFlex-Final-Outline>